Artificial Intelligence

VICTORIA UNIVERSITY OF WELLINGTON
TE HERENGA WAKA

AIML320/420
Ensemble Learning

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AIML320/420 Week 3

1. Announcements
   - Assignment 1 (**15%**)

     Due on 11:59pm,

     26th March (Thur), week 5
   - Helpdesk: Tuesday and Thursday, 2-3pm at CO242B
2. Algorithms Learned:
   - K nearest neighbour (KNN)
   - Decision Tree
3. Ensemble Learning

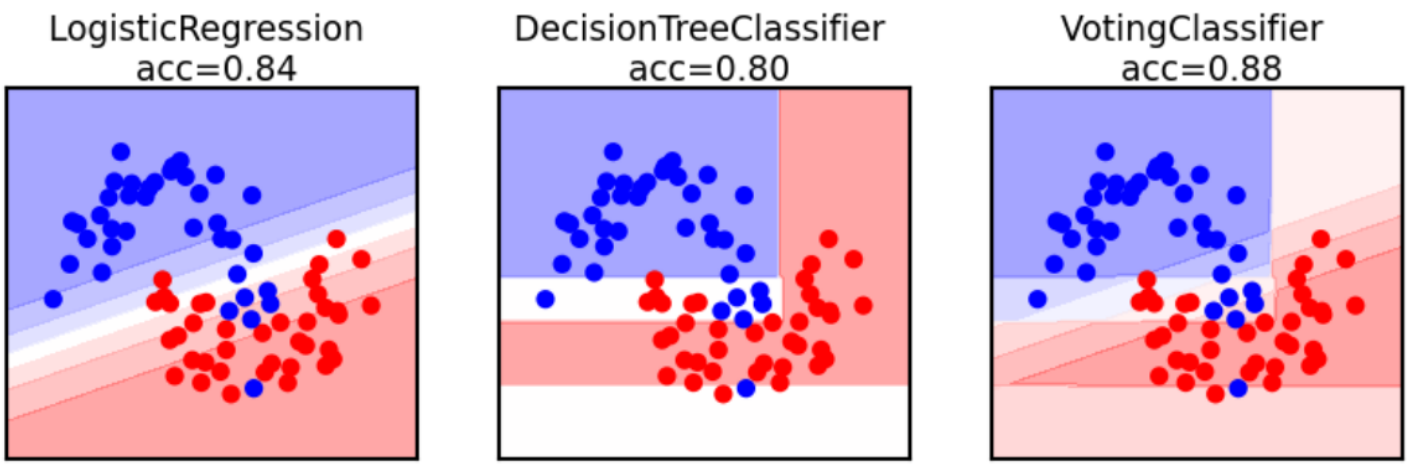
Outline

- Why ensembles - Intuitions
- Diversity, Combination and Base learners
- Algorithms
  - Bagging
  - Boosting, Adaboost
  - Random Forest

Why ensembles?

Instead of using a single learner, we train several learners and combine them… **why?**

- Can be the same or different types of learners



Why ensembles?

3 classifiers, each one of them is correct 6 out of 10
What happens when we combine their votes?

Ground-truth	c1	c2	c3
X	X	O	X
X	X	O	X
X	X	O	X
X	X	X	O
X	X	X	O
O	O	O	X
O	X	X	X
O	X	O	O
O	X	O	O
O	X	O	O

Why ensembles?

3 classifiers, each one of them is correct 6 out of 10
What happens when we combine their votes?

Ground-truth	c1	c2	c3	Maj Vote
X	X	O	X	X
X	X	O	X	X
X	X	O	X	X
X	X	X	O	X
X	X	X	O	X
O	O	O	X	O
O	X	X	X	X
O	X	O	O	O
O	X	O	O	O
O	X	O	O	O

Why ensembles?

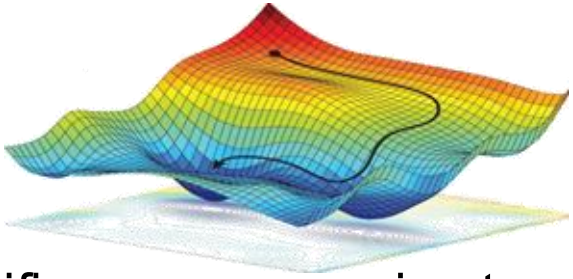
Better predictive performance

Intuitively, an ensemble should outperform any of its members individually in terms of predictive performance (e.g. accuracy)

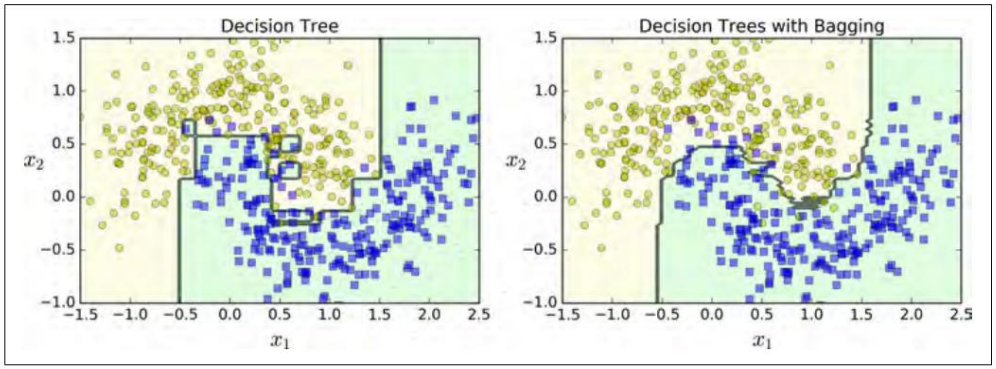
- Individuals often make mistakes, but the “majority” is less likely to make mistakes, mistake can be ‘averaged out’
- Individuals often have partial knowledge, but a committee can pool expertise to make better decisions.

Why ensembles?

- Imperfect training algorithm
  - Some classifiers may converge to a local minimum.
  - Combining them in a model even closer to the absolute minimum (optimal classifier) than any of them is capable individually.



- Several simple classifiers can approximate complex classification boundaries.



A single Decision Tree versus a bagging ensemble of 500 trees

Kuncheva, Ludmila I. Combining pattern classifiers: methods and algorithms. John Wiley & Sons, 2014.

Why ensembles- Some scenarios

- Too much data
  - Train many classifiers on small “subsets” of the data and aggregate their outputs.
- Too little data – reuse the data in a diverse way
  - There is just not enough data to build a “strong” classifier.
  - We can resort to **bootstrapping** to generate several subsets of data, train a classifier on each of them and obtain a better (combined) learner- reuse the data.
- Divide and conquer
  - The classification problem might be easier to solve if the data is split into smaller and easier-to-handle problems.
- Data fusion
  - The data may come from several different sources.

Three keys for Ensemble

Diversity, Combination, Base learner

Diversity: learners have to be “diverse” w.r.t their predictions

Combination: must suppress incorrect predictions and highlight correct predictions

Base learner: must be somewhat accurate

Inducing Diversity

Train on **different instances**: Bagging [1]

Train on **different features**: Random Subspaces [2]

Both: Random Forest [3] and Random Patches [4]

Heterogeneous learners: Combine different learners like decision trees, KNN, perceptrons, …

[1] Breiman, Leo. "Bagging predictors." *Machine learning*, Springer, 1996.

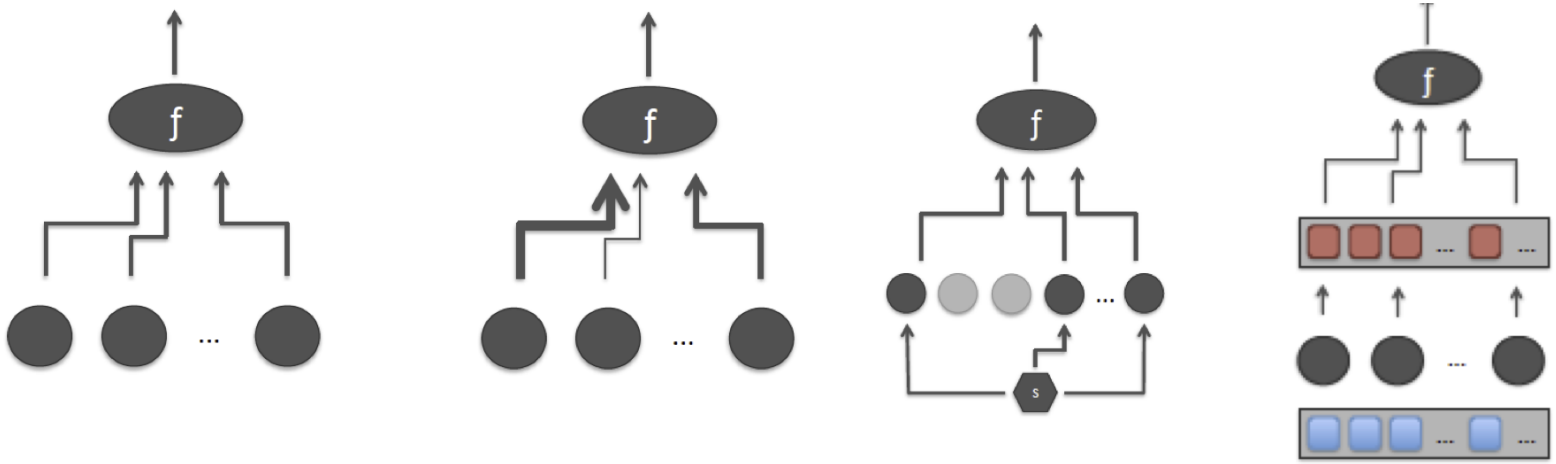
[2] Ho, Tin Kam. "The random subspace method for constructing decision forests." *IEEE TPAMI*, 1998

[3] Breiman, Leo. "Random forests." *Machine learning*, Springer, 2001

[4] Louppe, Gilles, and Pierre Geurts. "Ensembles on random patches." *ECML-PKDD 2012*

Combination

Defines how individual predictions are combined to form the ensemble prediction



Majority

choose whichever label is predicted by more than half of the learners

Weighted Majority

adds more weight on learners according to a weighting function (usually based on the predictive performance)

Classifier Selection

a function S maps input data to specific learners (others are not allowed to vote)

Meta-learning

Train a second-layer learner using the predictions of the first-layer learners (stacking)

[5] *Gomes, Heitor M., et al. "A survey on ensemble learning for data stream classification." ACM Computing Surveys (CSUR), 2017*

Base Learner

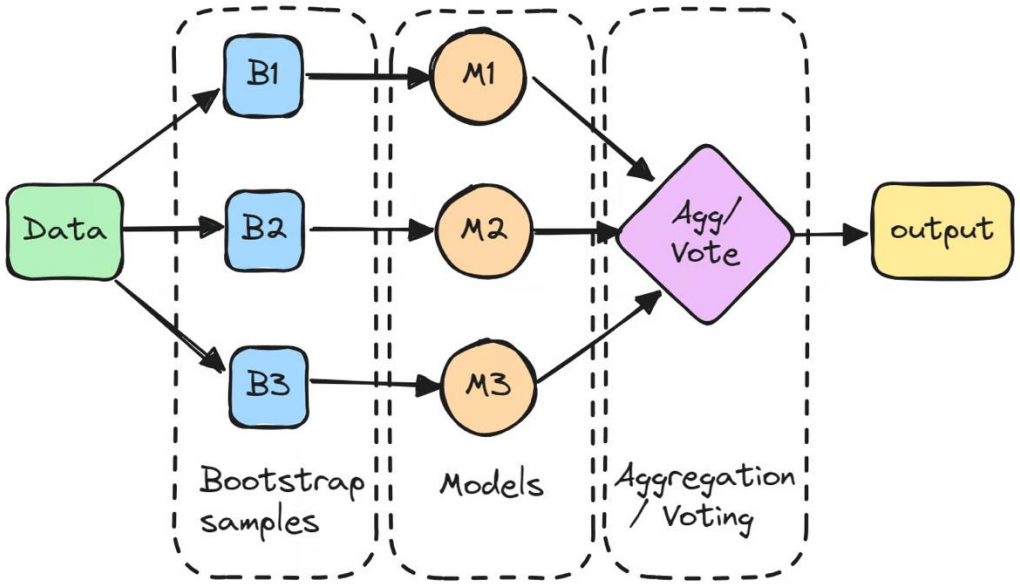
- **Individual accuracy**: at least better than random
- **Dependent / Independent**: the training of one learner depend on the other learners or not.
- "**Unstable**": given small perturbations the learning algorithm must generate different models
  - adaptive and collaborative

Diverse models + collaboration ---> better performance

Bagging: Bootstrap Aggregation

Bagging trains each model of the ensemble with a bootstrap sample from the original dataset

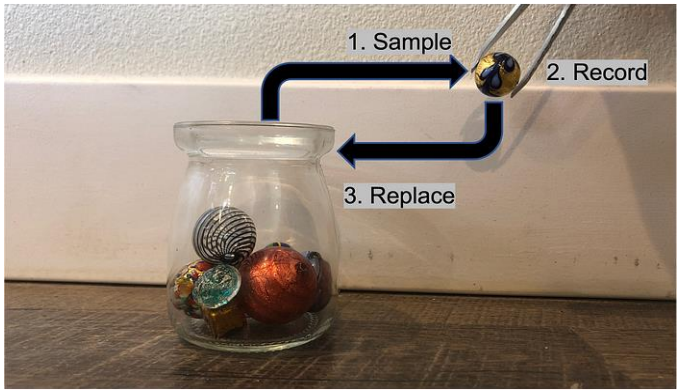
- a **random** subset of the training data sample **with replacement**



Breiman, Leo. "Bagging predictors." *Machine learning*, Springer, 1996.

Bagging– bootstrap sample

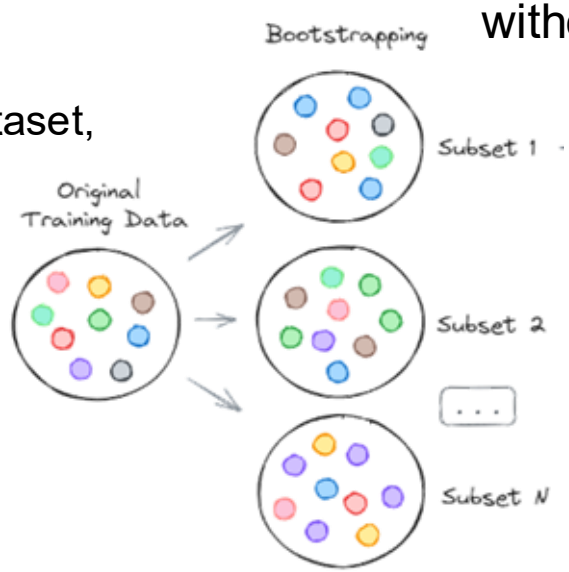
Sample **with replacement**: Sampling unit is returned before next is drawn



without replacement

In the sample with replacement:

- Some are drawn into new dataset, some are left out
- Some are repeatedly drawn



Breiman, Leo. "Bagging predictors." *Machine learning*, Springer, 1996.

Bootstrap Sampling

New datasets:

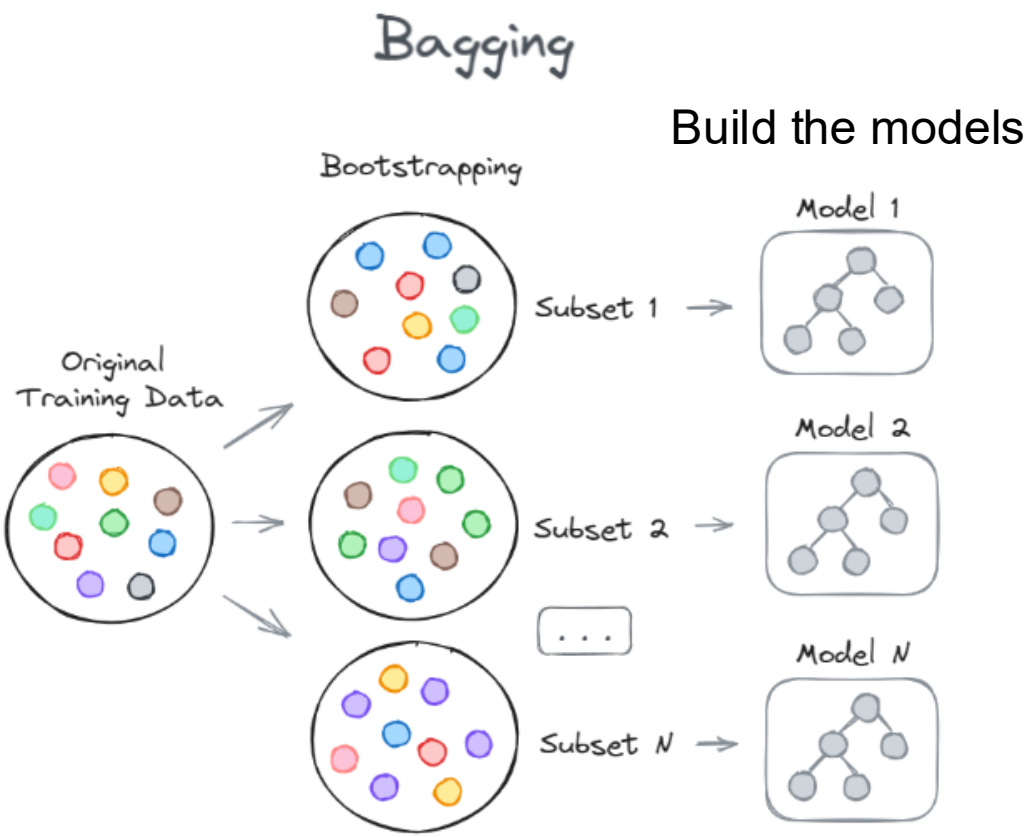
- Same size as original one
- For each one,
  - Some data is missing out
  - Some data occur more than once
  - Variance is reduced

Out-Of-Bag (OOB)

- Can be used to estimate the generalization error without a separate set (e.g., validation set)

Breiman, Leo. "Bagging predictors." *Machine learning*, Springer, 1996.

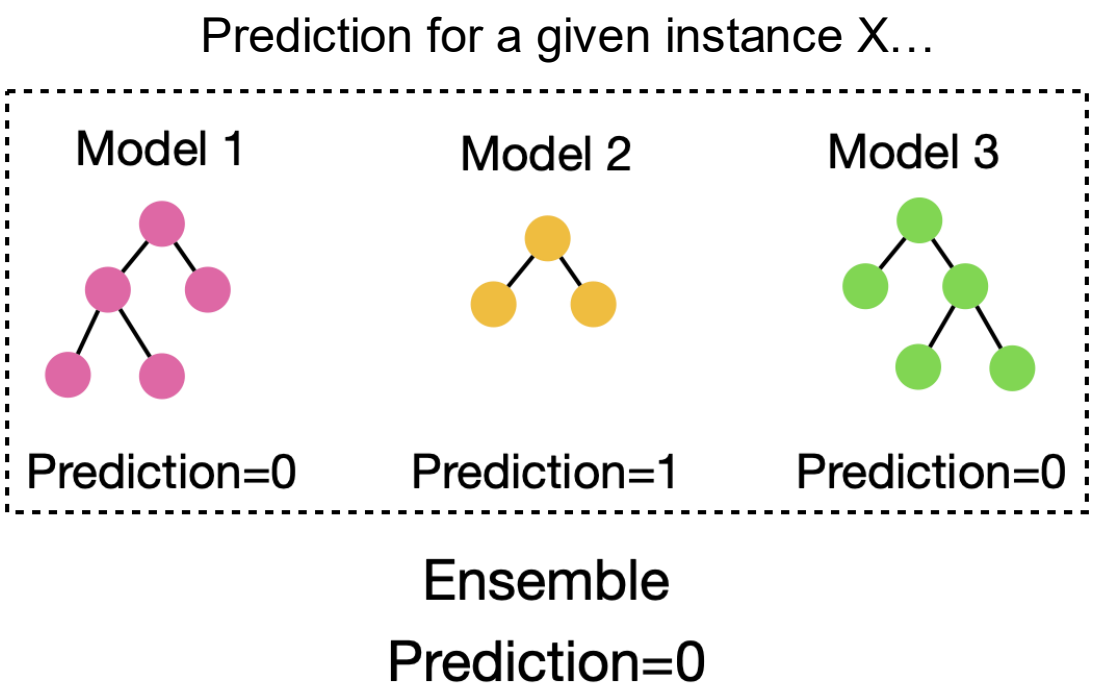
Bagging



Breiman, Leo. "Bagging predictors." *Machine learning*, Springer, 1996.

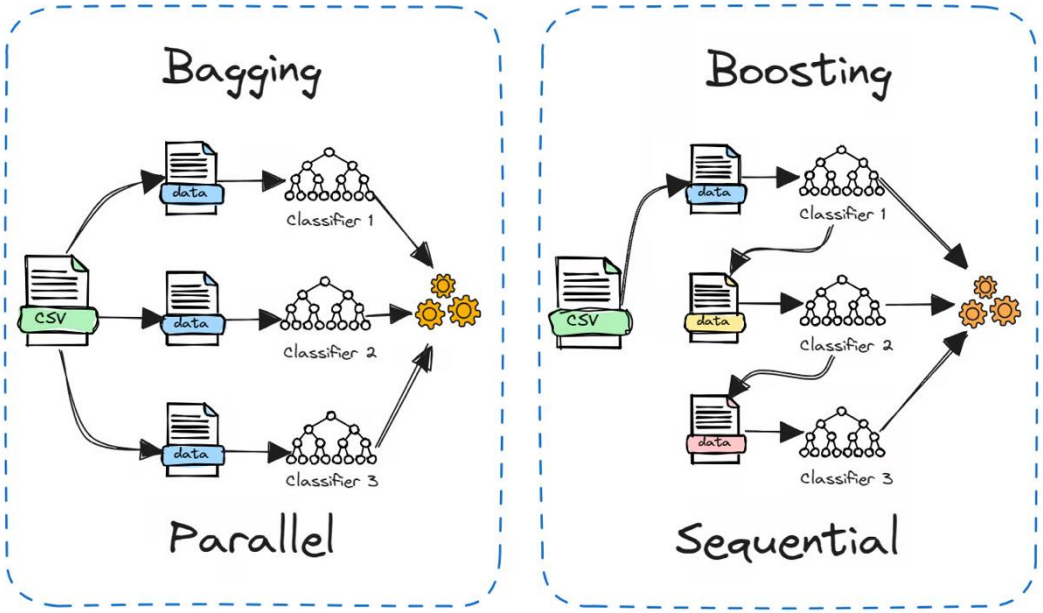
Bagging: prediction

The predictions of each learner are aggregated using majority vote to obtain the final prediction.



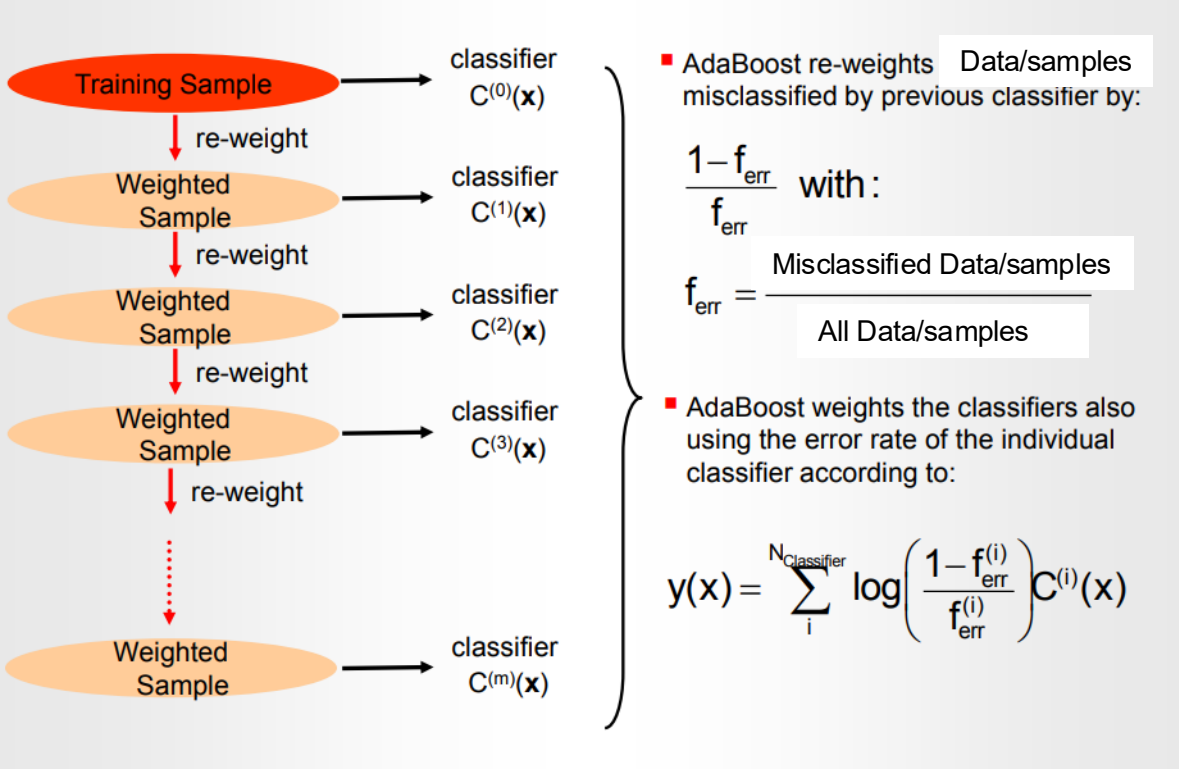
Boosting

- Boosting
  - Train classifiers sequentially, each time focusing on training examples/instances that the previous ones got wrong.
  - To focus on specific examples, boosting uses a weighted training set, i.e., a core mechanism where each data point is **assigned a specific weight** that is adjusted iteratively to force subsequent models to focus on the examples that previous models misclassified.



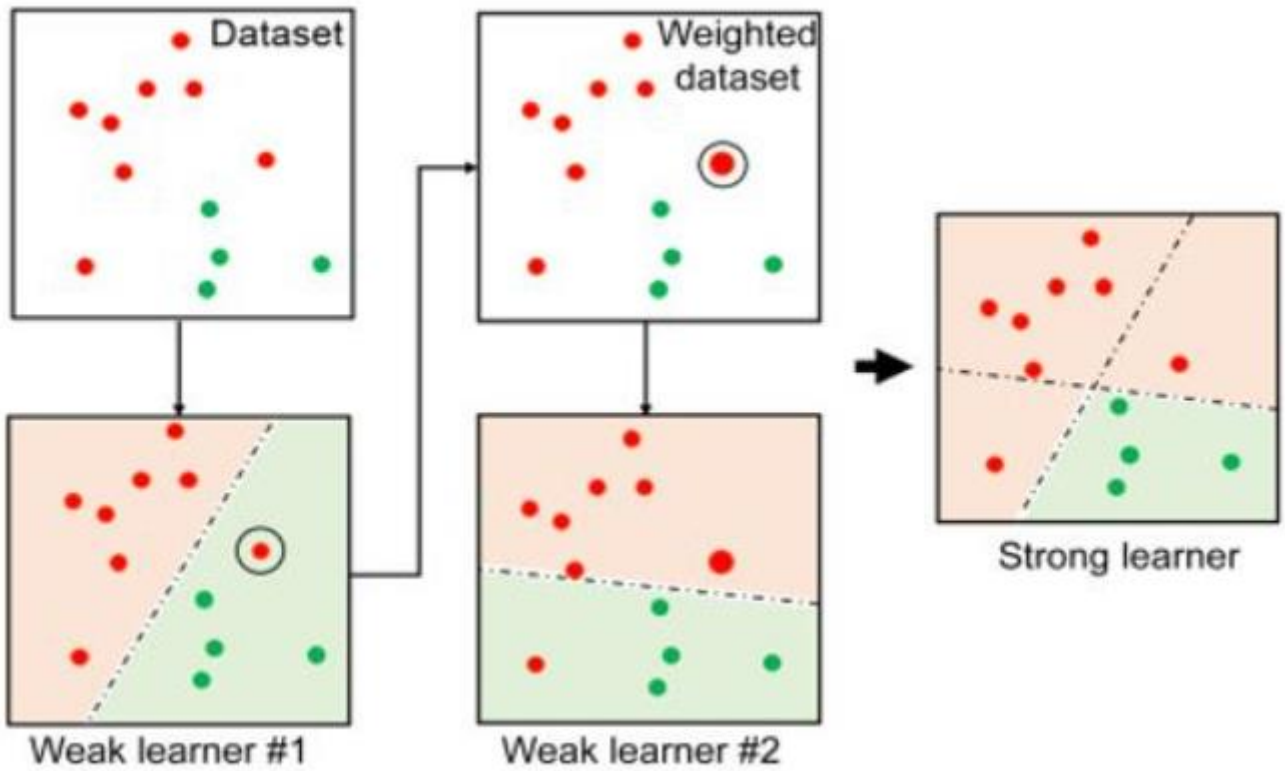
AdaBoost (Adaptive Boosting)

- Adaboost classifier is to increase the weight of the misclassified point, also decrease the weight of the classified points.



If a classifier has a smaller overall error, it will apply a much larger weight increase to the few instances it actually misclassified.

Illustration of AdaBoost

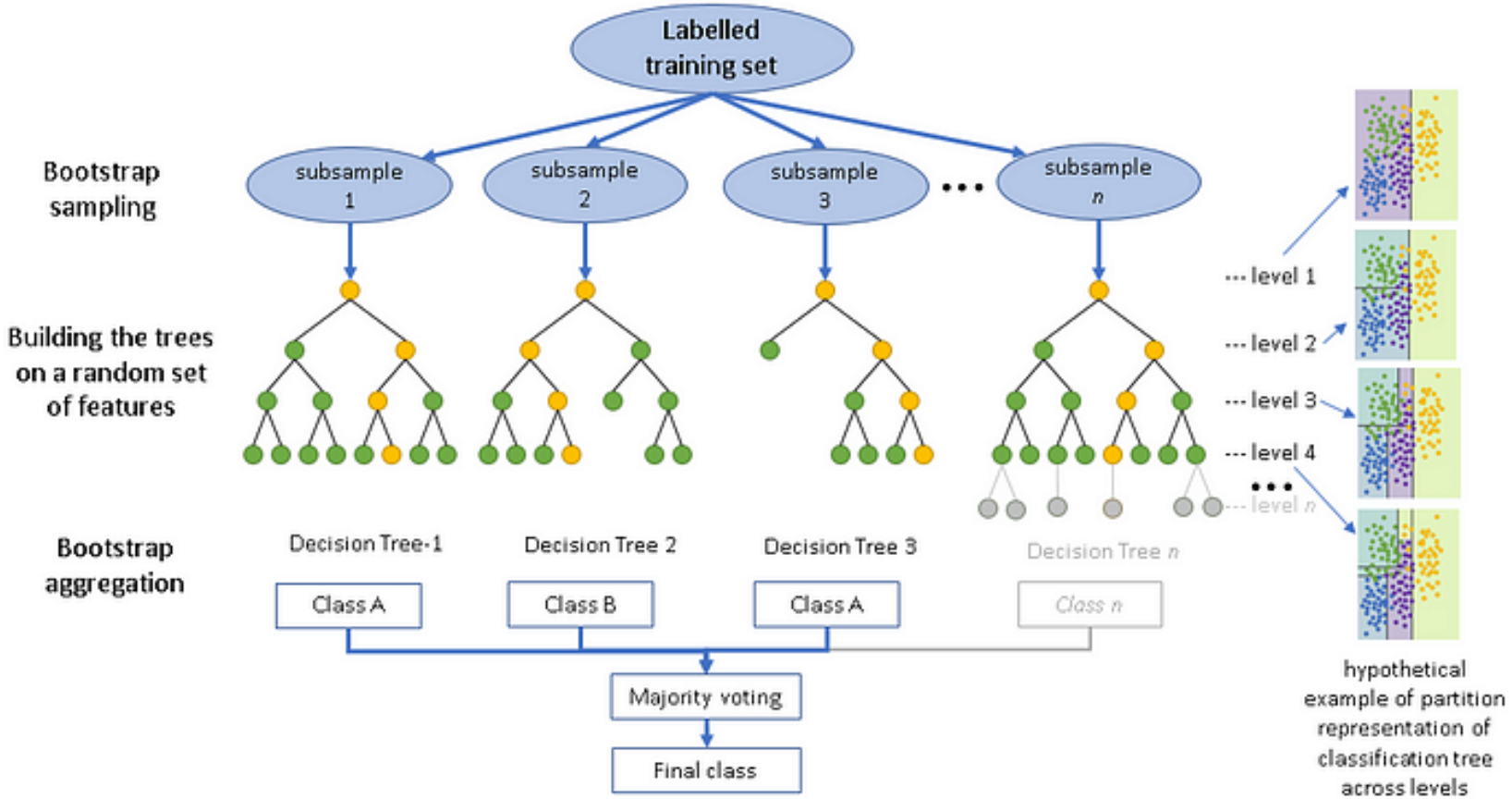


Random Forest

- Adds extra randomization on top of Bagging
  = Bagging with trees **+ random feature subsets**
- Base learner **must be a decision tree**

- Split decisions are made using **a random subset of features** selected

Random Forest



Random Forest

Hyperparameters (influential ones):

- **Number of trees**: The number of trees in the forest
- **Number of features**. The number of features to consider when looking for the best split
- **Number of samples**. The minimum number of samples required to split an internal node

Bias and Variance reduction. RFs are effective at reducing variance by introducing randomness during training and averaging predictions (combination)

Liu, Brian, and Rahul Mazumder. "Randomization Can Reduce Both Bias and Variance: A Case Study in Random Forests." arXiv preprint arXiv:2402.12668 (February 2024)

Summary

- Ensembles are powerful methods to improve predictive performance using several “weak learners”
- Diversity is a must, but the base learner and combination are important too
- Bagging, Adaboost and Random Forest are simple, yet powerful ensemble methods

Next Lecture:

- Data Preparation and Feature Engineering